

Representational Geometry in Neural Networks

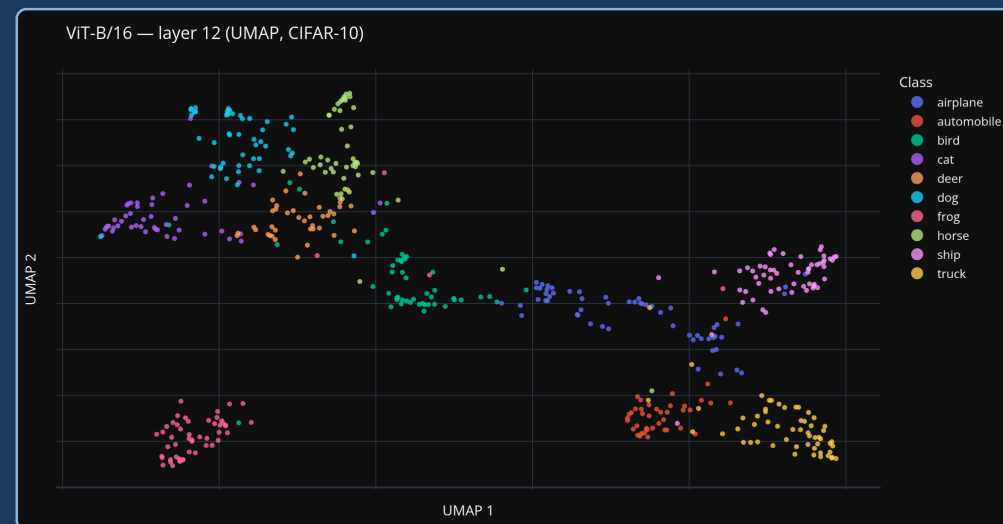
From Vision Transformers to RLHF-Aligned Language Models

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ViT-B/16 · layer 12 · CIFAR-10 · real data

Part 1 — Cross-modal alignment

When a vision model and a language model train on the **same task**, does a shared representational structure emerge across modalities?

- **Hypothesis A:** cross-modal signal is new information → improves model
- **Hypothesis B:** models converge regardless — extra signal adds little

CKA before/after cross-modal fine-tuning · DINOv2
+ Llama · MS-COCO

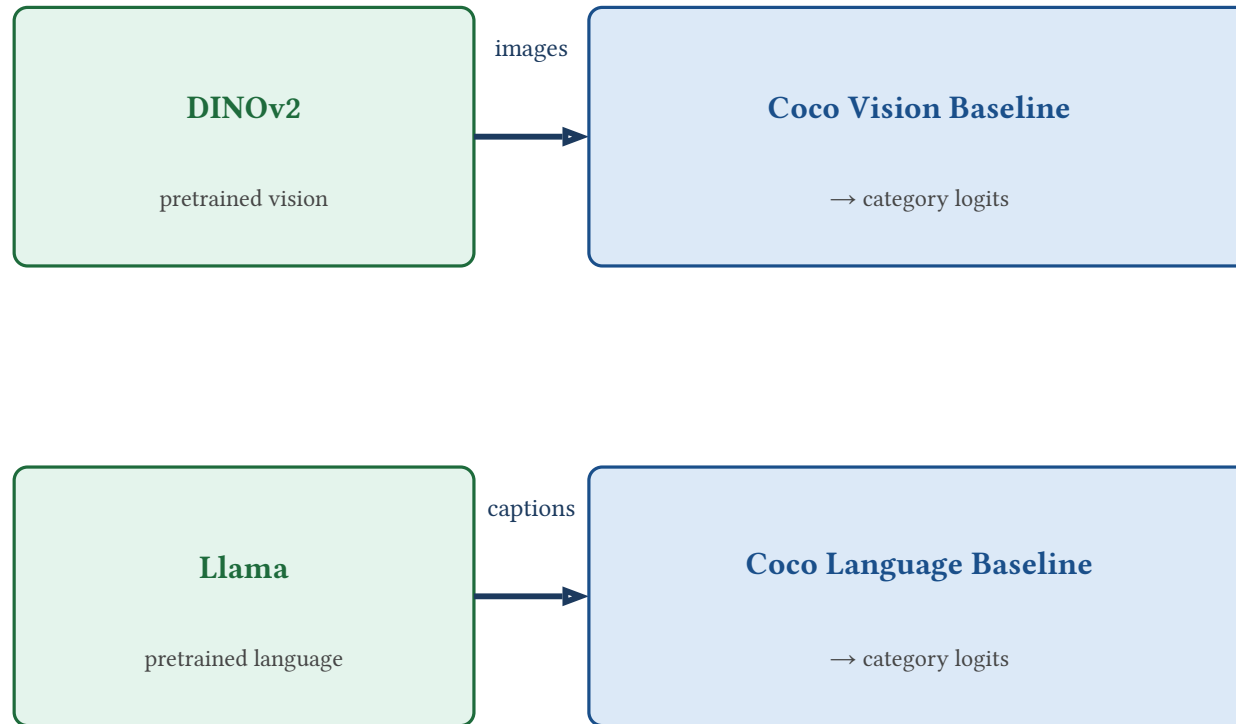
Part 2 — RLHF geometry

Does RLHF geometrically transform a language model's embedding space — encoding human preference as a **linearly separable structure**?

- Does alignment progressively separate chosen/rejected responses across SFT → DPO → RLHF?
- How does separation evolve **layer by layer**?

UMAP + LinearSVC per layer · Tulu-3-8B SFT/DPO/
RLHF · Anthropic/hh-rlhf

Part 1 — Setup: Two Independent Baselines on MS-COCO



Part 1 — Example: MS-COCO 2017

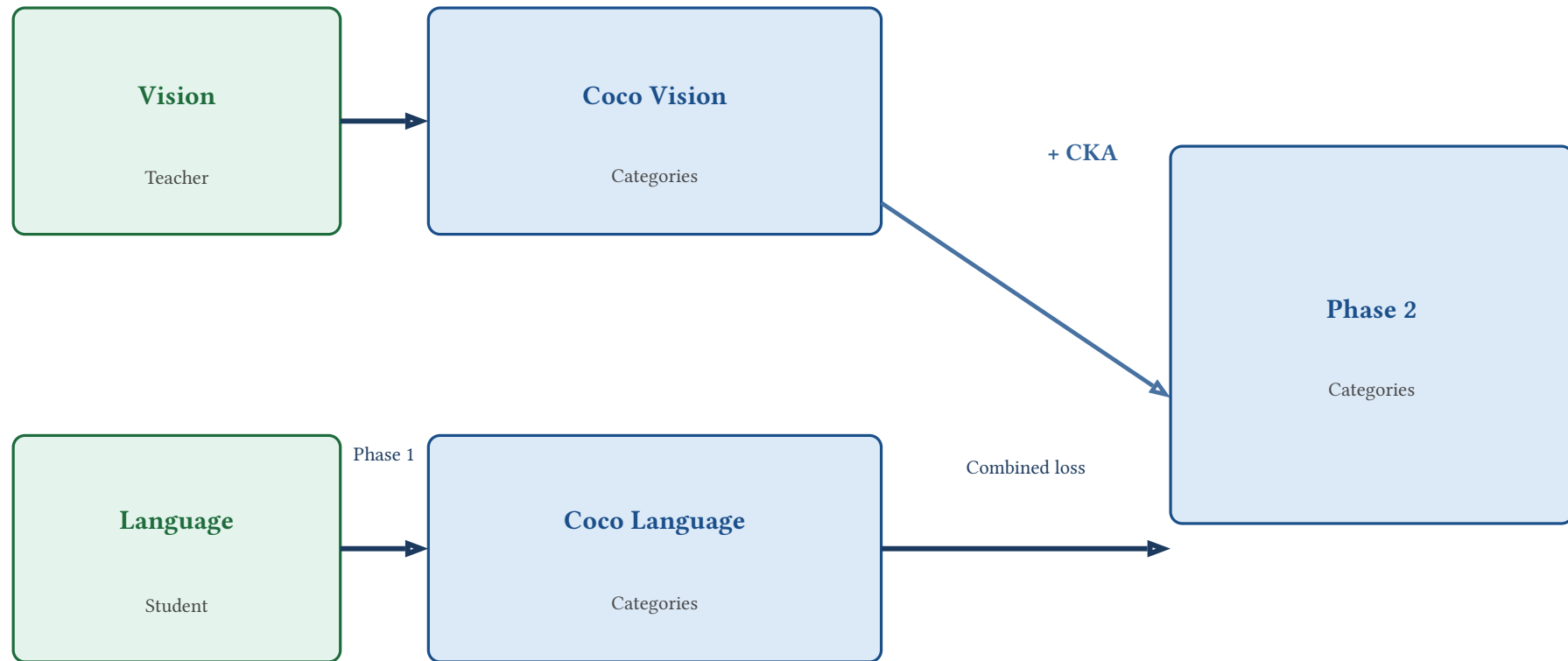
Categories: person, tie, umbrella, handbag

Captions:

- A girl in grey jacket and tie standing on a street.
- Little girl posing for the camera with an adult sized striped tie on.
- A kid with a tie standing in the street.
- A young girl wearing a necktie and smiling broadly.
- Girl smiles for picture in busy Asian plaza.



Part 1 — Method: Teacher → Student CKA Fine-tuning



Part 2 — Motivation: Why RLHF?

Pretraining and SFT optimize **likelihood** — not **preference**.

Without RLHF

- Next-token prediction rewards **plausible** text
- No signal for helpful / harmless / honest
- A great answer and a subtly wrong one can look equally likely

With RLHF

- Optimizes directly for **human preference**
- Signal comes from comparisons, not a fixed label set
- Model improves on axes humans care about, not just fluency

Insight: Goal Shift probability mass toward responses humans rank higher — without collapsing fluency or drifting into reward hacking.

Part 2 — Three Steps of RLHF

Classic formulation (Ouyang et al., 2022 · InstructGPT) — the paradigm Part 2 measures the geometric effect of

- ## 1. Collect preference data

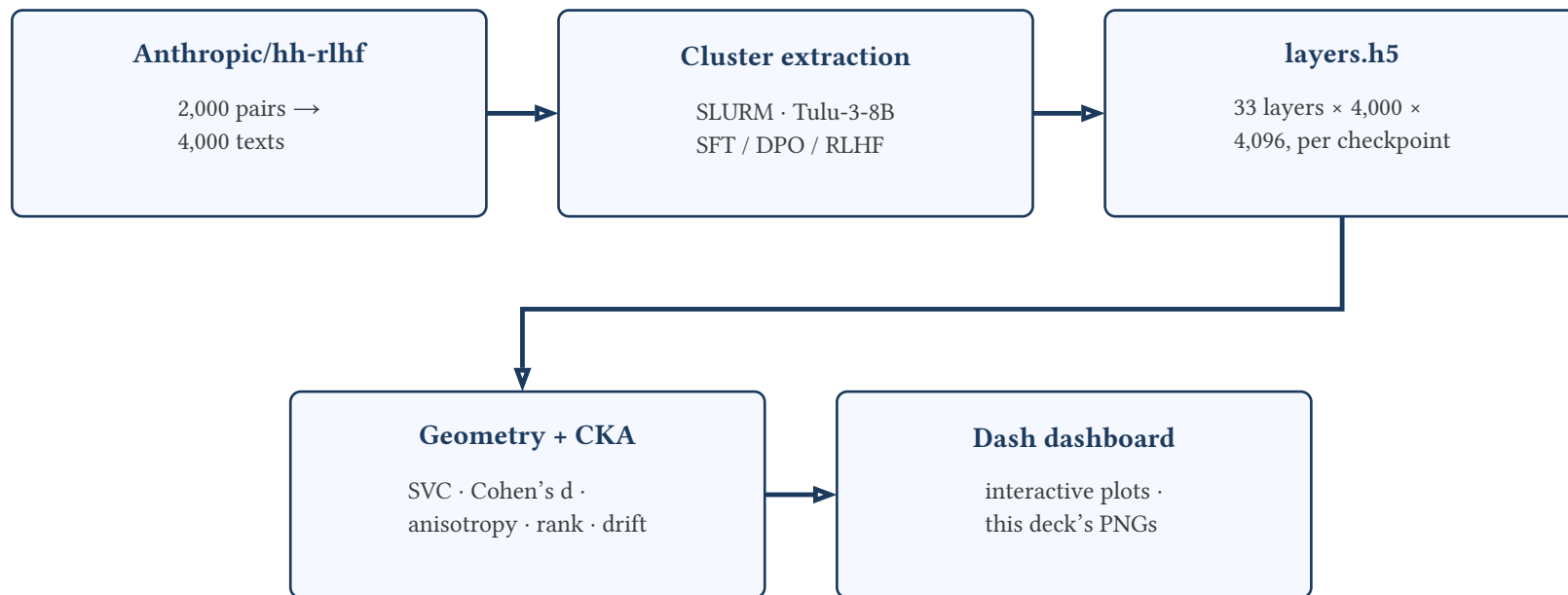
 - Sample **multiple** completions per prompt from the SFT model
 - Humans compare pairs $\rightarrow$ chosen y_w , rejected y_l
 - Output: dataset of (x, y_w, y_l) triples
- ## 2. Train a reward model

 - $r_\varphi(x, y)$ learns to **score** a completion
 - Bradley–Terry pairwise loss:
$$\mathcal{L}(\varphi) = -\mathbb{E}[\log \sigma(r_\varphi(x, y_w) - r_\varphi(x, y_l))]$$
 - Learns to rank, not to generate
- ## 3. Policy optimization

 - π_θ generates, r_φ scores, PPO updates θ
$$\max_{\theta} \mathbb{E}_{y \sim \pi_\theta} [r_\varphi(x, y)] - \beta \text{KL}(\pi_\theta \parallel \pi_{\text{ref}})$$
 - KL term stops **reward hacking**

Remark Our third checkpoint (Llama-3.1-Tulu-3-8B) is **not** this pipeline. Tulu-3 replaces steps 2–3 with **RLVR** (Reinforcement Learning with Verifiable Rewards) — rule-based, verifiable rewards (math/code correctness, constraint checks), no learned reward model at all. We call it “RLHF” only for consistency with Part 1’s vocabulary. The **DPO** checkpoint skips the reward model differently — it optimizes directly on the implicit log-ratio $\beta \log \left(\frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)} \right)$ between chosen and rejected, with no reward model or RL loop at all.

Part 2 — Pipeline: From Dataset to Dashboard



`scripts/extract_embeddings.py` → `app/rlhf_geometry_compute.py` /
`app/vision_compute.py` → `app/app.py`

Part 2 — What Each Checkpoint Actually Is

Tulu-3-8B alignment trajectory — 3 public AllenAI checkpoints, same base model, extracted at each post-training stage

SFT

Llama-3.1-Tulu-3-8B-SFT

- Instruction fine-tuning
- No preference signal yet

DPO

Llama-3.1-Tulu-3-8B-DPO

- Chosen > rejected, directly
- No reward model

RLHF

Llama-3.1-Tulu-3-8B

- Final RL stage (Tulu-3's RLVR)
- Labeled **RLHF** for Part 1 consistency

Remark All 3 are the **same** underlying 8B model at 3 successive training stages — not 3 different architectures. This is what makes the SFT → DPO → RLHF comparison a clean trajectory rather than a cross-model comparison.

Definition: Linear CKA For column-centered X, Y :

$$\text{CKA}(X, Y) = \frac{\|X^\top Y\|_F^2}{\|X^\top X\|_F \cdot \|Y^\top Y\|_F}$$

1 = proportional representations (no drift), 0 = orthogonal.

Definition: Anisotropy Over randomly sampled pairs $i \neq j$:

$$\text{aniso}(X) = \mathbb{E}_{i \neq j} [\cos(x_i, x_j)]$$

Near 1 = representations collapse to a narrow cone.

Definition: Cohen's d Direction $v = \frac{\mu_{\text{chosen}} - \mu_{\text{rejected}}}{\|\mu_{\text{chosen}} - \mu_{\text{rejected}}\|}$:

$$d = \frac{(\mu_{\text{chosen}} - \mu_{\text{rejected}}) \cdot v}{\sigma_{\text{pooled}}}$$

Effect size along the best linear separating direction.

Definition: Effective rank Participation ratio of covariance eigenvalues λ_i :

$$\text{erank}(X) = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}$$

How many dimensions are effectively in use.

Tulu-3-8B SFT → DPO → RLHF · Anthropic/hh-rlhf · 4,000 samples · 33 layers

Negative finding

- Not linearly encoded, any stage
- LinearSVC peak: 0.543 / 0.544 / 0.550
- Near chance, all 3 checkpoints (layer 14)
- Cohen's d peak: 0.183 → 0.193 → 0.194
- No phase transition in preference geometry

But the geometry does shift

- Linear CKA, same inputs, layer by layer
- CKA(SFT, DPO) drops to **0.976** by layer 32
- Most of the shift happens SFT→DPO
- CKA(DPO, RLHF) stays near **0.999**
- RLHF/PPO barely moves geometry further

Full curves in the dashboard · Part 2 tab

Remark Peak accuracy near 0.55 matches published hh-rlhf baselines (0.57–0.66) and the dataset's own 63% label-agreement ceiling (Bai et al., 2022).

Part 2 — How Deep Can a Probe Even Go Here?

Reading the 0.55 baseline before the flat curves show up

What caps the accuracy?

- Bai et al. (2022): 63% researcher–crowdworker agreement
- Stiennon et al. (2020): $77\% \pm 2\%$ with careful onboarding
- Label quality is a dataset property, not a constant
- Reward models on hh-rlhf: 57–66%, rarely above 72%

What that means for our 0.55

- Can't beat the label consistency of the data
- 0.55 is low, not far outside published baselines
- Real finding: the **missing layer trend**
- Not the raw accuracy number itself

Sources: Bai et al. 2022 · Stiennon et al. 2020 · reward-model benchmarks on hh-rlhf

Part 2 — “No Linear Separation”: an RLHF Finding, or a DPO Finding?

Six objectives, side by side

- Sinha et al. (2026, arXiv:2606.09850): layer-wise probing, 6 objectives
- **KTO**, **GRPO** build linear separability (constructive)
- **DPO**, **ORPO** degrade it (geometric rotation)
- **PPO**, **SimPO** largely preserve baseline geometry
- Our RLHF checkpoint $\approx$ RLVR/PPO-like $\rightarrow$ fits “preserves” pattern
- Our DPO step: biggest CKA jump, **no** separability gain
- Fits “degrades via rotation,” not “constructs a new axis”

Remark Our negative finding is likely specific to DPO/PPO-style objectives — not to “RLHF” in general.

Source: Sinha et al. 2026 (arXiv:2606.09850) — very recent preprint, re-check before citation

Part 2 — Methodology Check: Pooling Decides a Lot

Approach	What's measured	Typical accuracy
Single-point probe (our approach)	Chosen/rejected as independent points, last-token pooled	0.5–0.6
Paired-difference probe	chosen – rejected per prompt pair, removes prompt variance	0.84–0.86 (arXiv:2604.09870)

- Our LinearSVC fights prompt-to-prompt variance
- Paired-difference removes that variance
- Not a weakness — a deliberately stricter question
- “Readable in raw space?” vs. “readable given pairs?”

Remark Next step: re-run a paired-difference probe as a robustness check to see whether separability rises once prompt variance is removed.

Source: Kirin 2026 (arXiv:2604.09870) — very recent preprint, re-check before citation

Independent cross-domain evidence for “SFT is the real turning point”

Language, layer-wise SFT

- `arXiv:2604.11838`
- Final layers far more sensitive to SFT
- Early/middle layers stay stable
- Matches our SFT→DPO jump

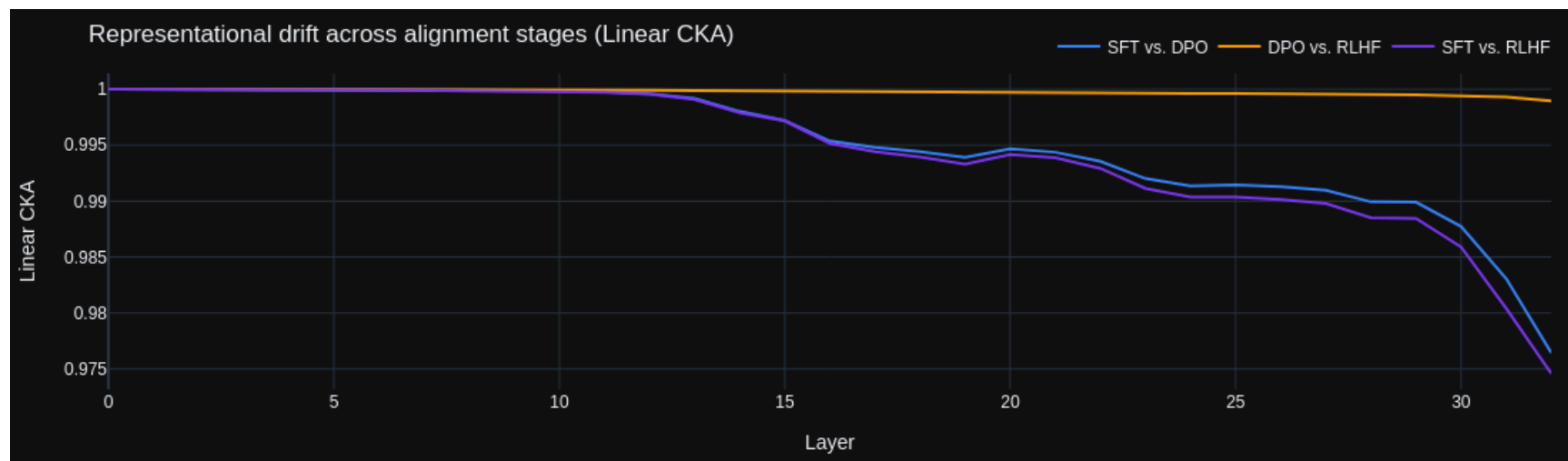
Speech domain, GRPO vs. SFT

- `arXiv:2607.08409`
- GRPO changes behavior, not representations
- Late RL step $\approx$ small geometry change
- Consistent with our DPO→RLHF pattern

Our “SFT→DPO is the turning point” finding rhymes elsewhere — strengthens the interpretation.

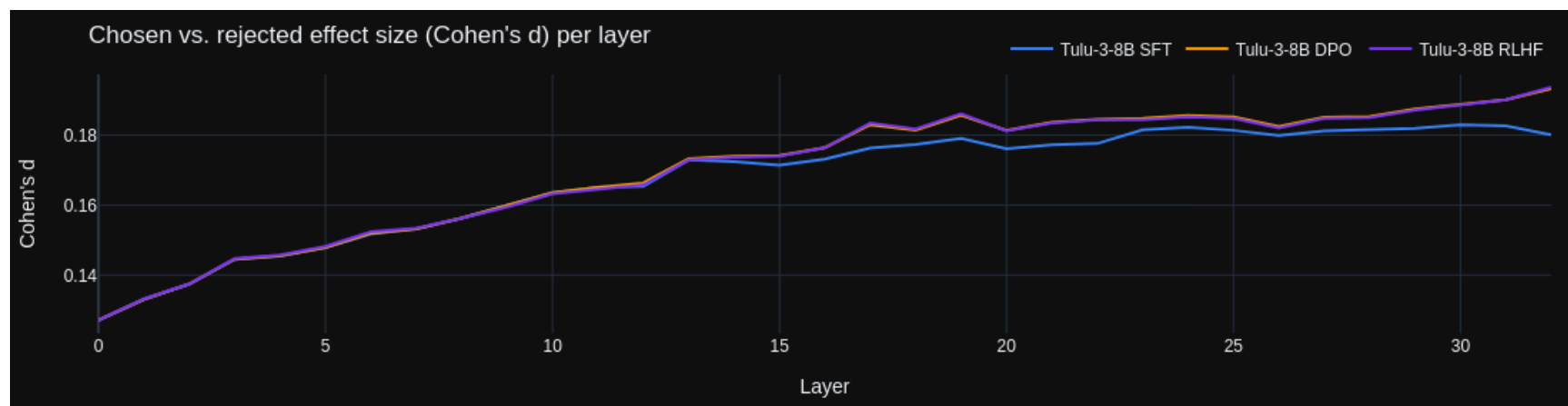
Sources: Zhao et al. 2026 (`arXiv:2604.11838`) · Kumar et al. 2026 (`arXiv:2607.08409`) — very recent preprints, re-check before citation

Part 2 — CKA Drift



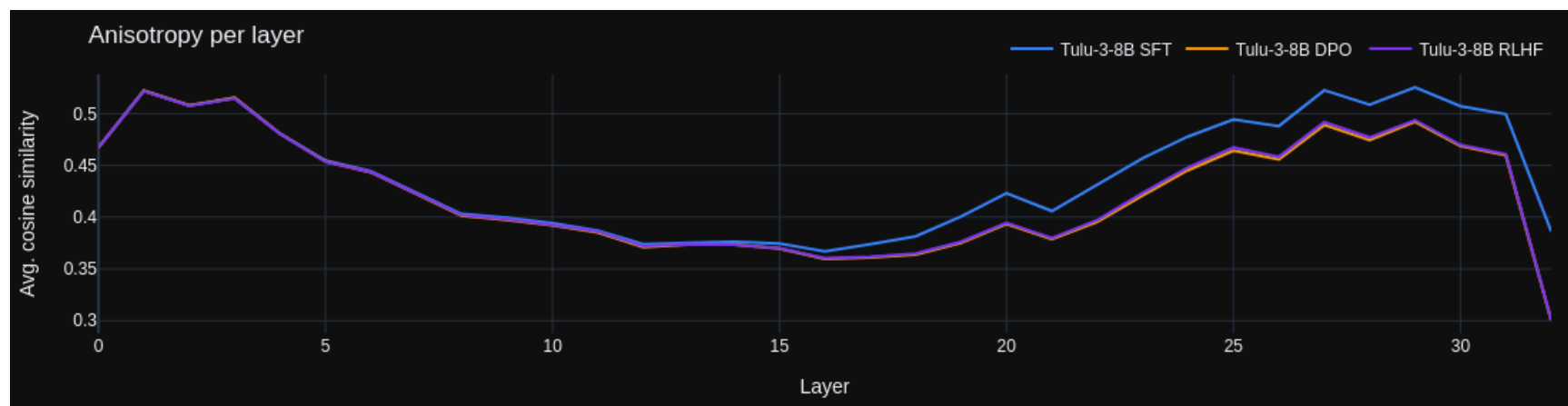
CKA drift — shift concentrated SFT to DPO, later layers

Part 2 — Cohen's d



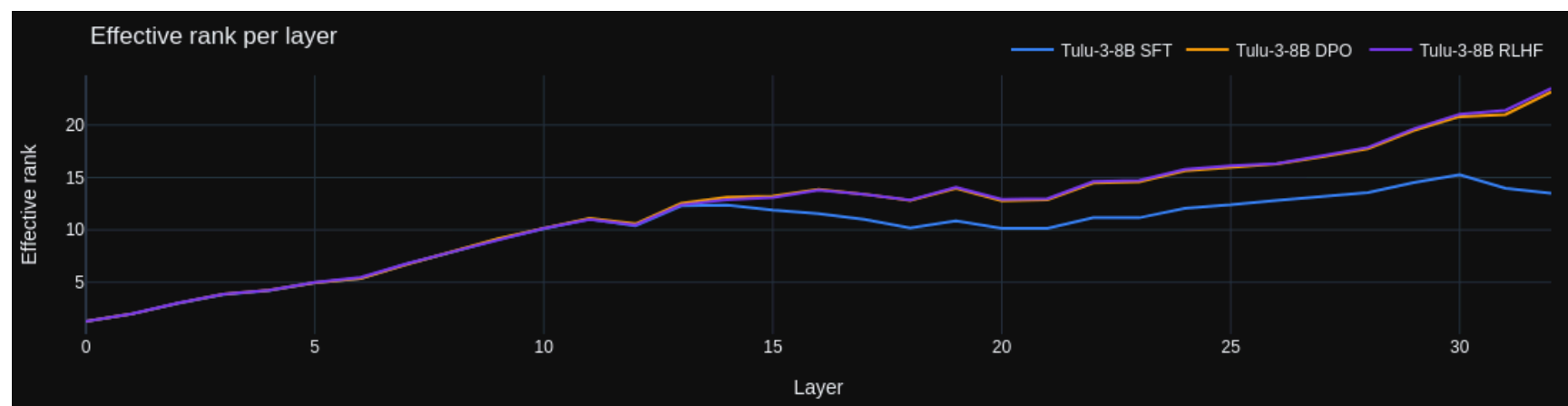
Cohen's d — small, flat across all layers

Part 2 — Anisotropy



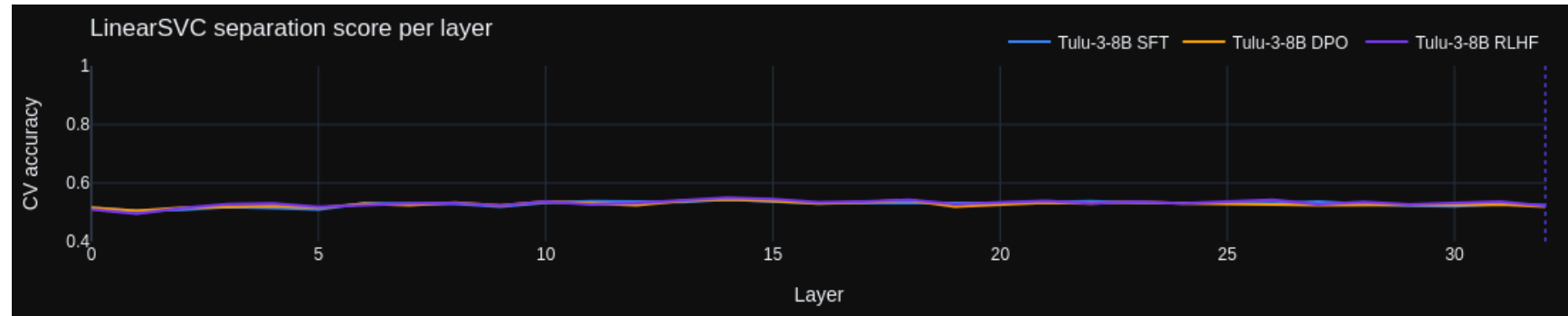
Anisotropy — dips mid-stack, rises again late

Part 2 — Effective Rank



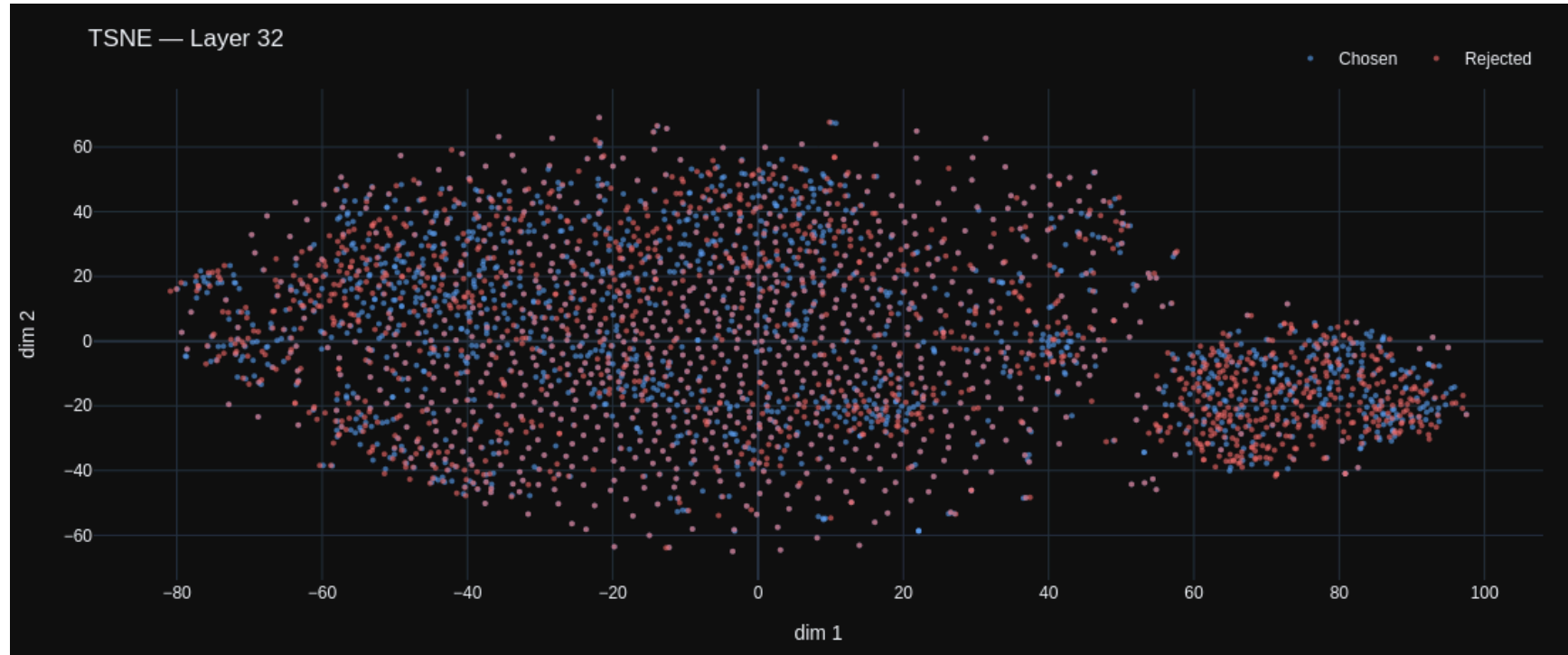
Effective rank — dimensionality usage per layer

Part 2 — LinearSVC Accuracy



LinearSVC accuracy — pinned near chance (0.5)

Part 2 — t-SNE Layer 32



t-SNE, layer 32 — chosen/rejected heavily overlap

Insight: no readable linear “preference direction”

- Flat, near-chance at **every** layer — all 3 checkpoints
- Peak SVC: 0.543 / 0.544 / 0.550
- DPO/PPO preserve or degrade separability; KTO/GRPO construct it
- Looks objective-specific, not universal to “RLHF”

Insight: but geometry is not static

- Real structural reorganization, concentrated at **SFT → DPO**
- CKA(SFT, DPO) drops to 0.976; CKA(DPO, RLHF) stays near 0.999
- Most “alignment work” happens before PPO

Remark Anisotropy and effective rank shift in that same window — one transition, not three.

*Alignment reshapes geometry **globally**, not a single axis. Lee et al. (2024): DPO **bypasses** a capability rather than erasing it.*

Part 1 — Cross-modal Alignment · Outlook

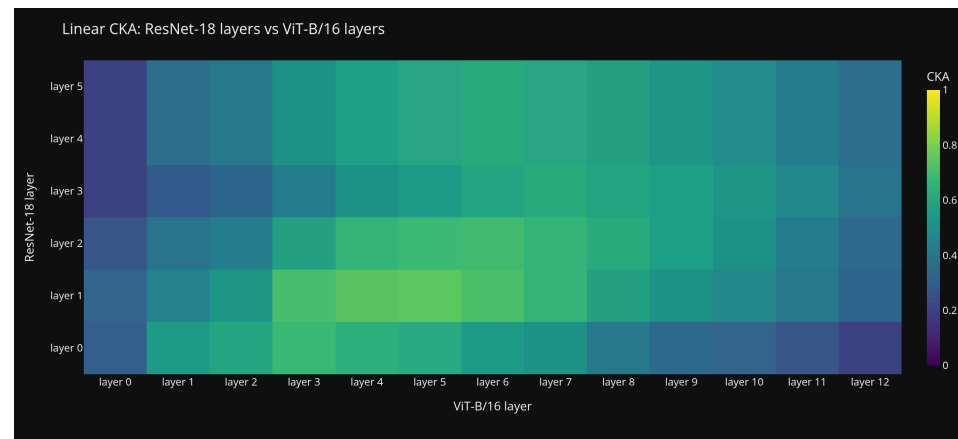
Setup · DINOv2 + Llama · MS-COCO · multi-class prediction

Both models trained on **same task** (object category prediction) · measure CKA · fine-tune with loss term encouraging similarity to the other model's representation · compare

CKA high before fine-tuning: models converge inherently (**Platonic Representation Hypothesis**)

CKA rises after fine-tuning: cross-modal signal is genuinely new information

Layer-wise CKA map shows **where** shared meaning lives · UMAP + dashboard to visualize

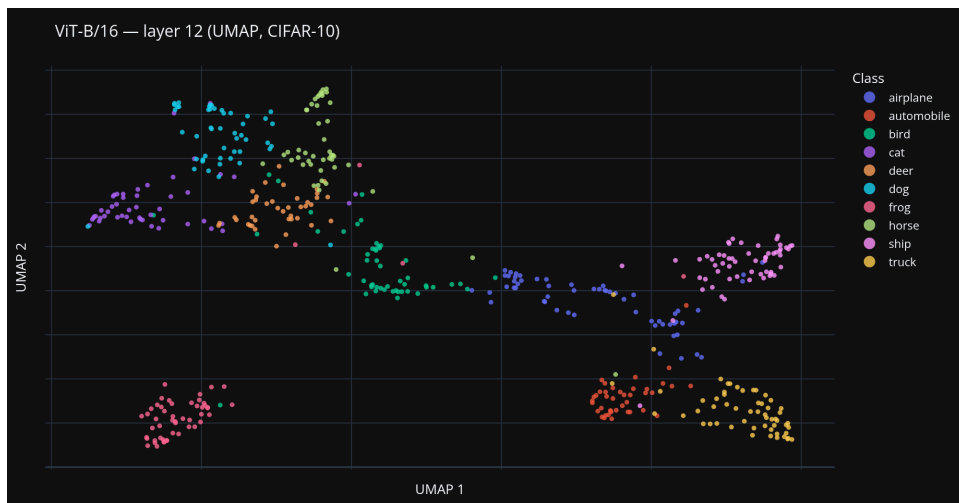


Interactive Plotly Dash · CKA heatmap

Next steps — toward final

- Cross-modal CKA: DINOv2 vs. Llama on MS-COCO pairs
- **CLIP (ViT-B/32)** as upper bound — explicitly trained cross-modal alignment
- Compare base vs. Instruct CKA scores

Deliverable & Next Steps



Interactive Plotly Dash · CKA heatmap · UMAP layer slider · LinearSVC score

Project deliverable

The dashboard **is** the deliverable —
interactive visualization of all findings:
layer slider, CKA heatmaps, UMAP scatter,
SVC score

Perspective

Extend to RLHF training **snapshots** —
plot cross-modal CKA vs. training step to
see if alignment shifts **during** RLHF

He, Trott, Khosla (2025)

Shared Latent Representations across Vision and Language

arXiv:2509.20751 · **Anchor paper**

Kornblith et al. (2019)

Similarity of Neural Network Representations Revisited

ICML · arXiv:1905.00414 · **CKA method**

Christiano et al. (2017)

Deep RL from Human Preferences

NeurIPS · arXiv:1706.03741 · **RLHF foundations**

Lambert et al. (2024)

Tulu 3: Pushing Frontiers in Open Language Model Post-Training

arXiv:2411.15124 · **Models used for Part 2**

Kucukahmetler et al. (2026)

Relative Geometry of Neural Forecasters

TMLR · arXiv:2602.15676

Ouyang et al. (2022)

Training LMs to Follow Instructions with Human Feedback

NeurIPS · arXiv:2203.02155 · **InstructGPT / RLHF**

McInnes et al. (2018)

UMAP: Uniform Manifold Approximation and Projection

arXiv:1802.03426 · **Dim. reduction method**

Huh, Cheung, Wang, Isola (2024)

Position: The Platonic Representation Hypothesis

ICML · arXiv:2405.07987 · **“Models converge inherently” framing**

References — Part 2 (RLHF Geometry)

Bai et al. (2022)

Training a Helpful and Harmless Assistant with RLHF
arXiv:2204.05862 · **Original hh-rlhf dataset paper**

Sinha, Garg, Elluru, Singh, Garg (2026)

Mechanistic Analysis of Alignment Algorithms in Language Models
arXiv:2606.09850 · **Objective-dependent separability · re-check**
Stiennon et al. (2020)

Learning to Summarize from Human Feedback
NeurIPS · arXiv:2009.01325 · **77%±2% agreement, contrast to hh-rlhf**
Zhao, Gong, Chen, Kang, Li (2026)
A Layer-wise Analysis of Supervised Fine-Tuning
arXiv:2604.11838 · **Final layers most SFT-sensitive · re-check**

Lee, Bai, Pres, Wattenberg, Kummerfeld, Mihalcea (2024)

A Mechanistic Understanding of Alignment Algorithms: A Case Study on DPO and Toxicity
ICML · arXiv:2401.01967 · **DPO learns an “offset”, not erasure**

Ziegler et al. (2019)

Fine-Tuning Language Models from Human Preferences
arXiv:1909.08593 · **Labeler agreement baselines**
Kirin (2026)

Relational Preference Encoding in Looped Transformer Internal States
arXiv:2604.09870 · **Paired-diff probes reach 84.5% · re-check**
Kumar et al. (2026)
When Synthetic Speech Is All You Have: Better Call GRPO
arXiv:2607.08409 · **GRPO changes behavior, not geometry · re-check**